

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

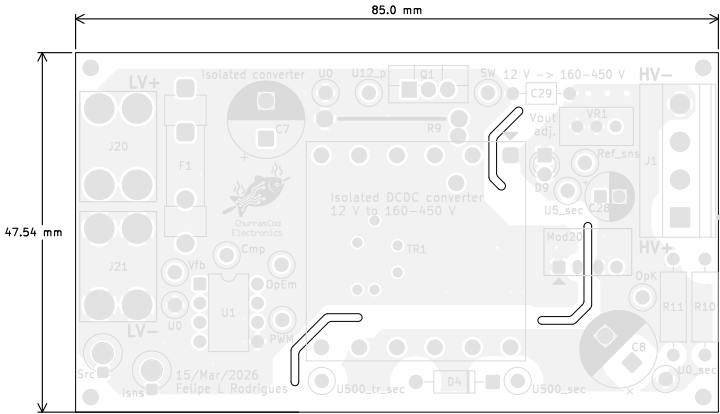
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL LEAD-FREE
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BUYER.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

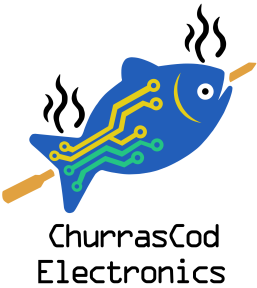
BOARD SIZE	85,000 × 47,540 mm
BOARD THICKNESS	1,600 mm
TRACE WIDTH	0,300 mm
TRACE TO TRACE	0,020 mm
MIN. HOLE (PTH)	0,250 mm
MIN. HOLE (NPTH)	2,100 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,200 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,254 mm



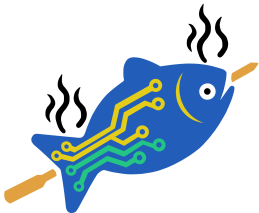
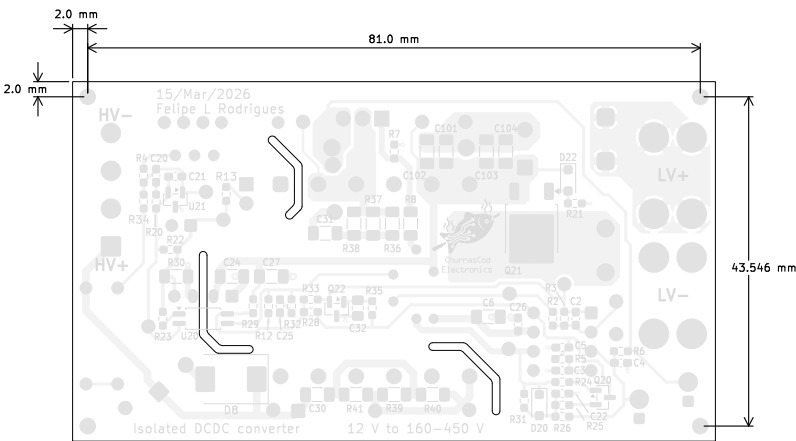
Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F,Mask		0,02 mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
Core		1,49 mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
B,Mask		0,02 mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,6 mm
Note: external layer thicknesses are specified after plating



Comments: Sheet Title: Top Fabrication (Scale 1:1) Sheet Path:	Company: NA		Variant: CHECKED		Git Hash: 9653abd
	Board Name: Step-Up			Project Name: Isolated step-up 12:450 V	
	File Name: Isolated_12to450V.kicad_pcb		Designer: FR	Date: 2026-02-07	Revision: 3.0.0+ (Unreleased)
	Reviewer: FR		Size: A4	Sheet: 1 of 10	



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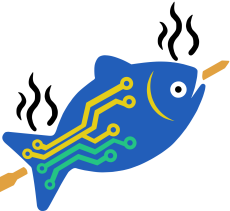
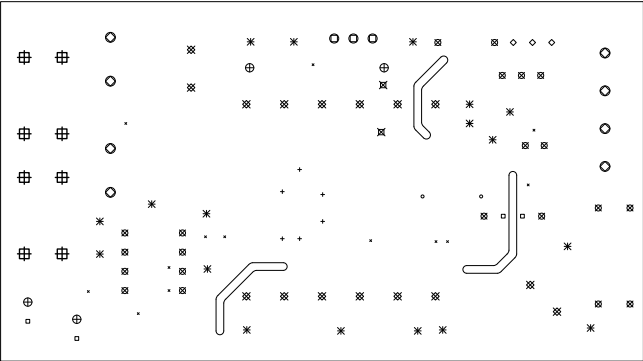
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Sheet Path:			Reviewer: FR	Revision: 3.0.0+ (Unreleased)
			Size: A4	Sheet: 2 of 10

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0,40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
+	6	0,50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
□	4	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	3	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	21	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
✱	18	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊗	2	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	16	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	4	1,10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 108						



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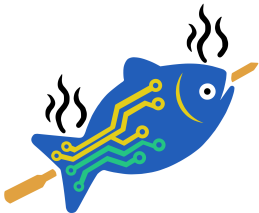
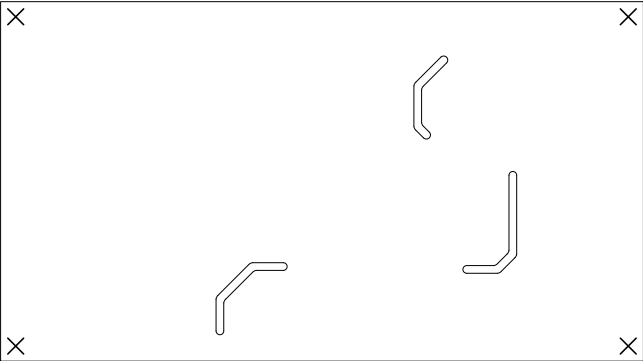
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Sheet Path:		Reviewer: FR	Size: A4	Sheet: 3 of 10

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					

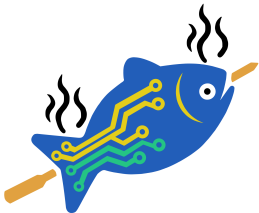
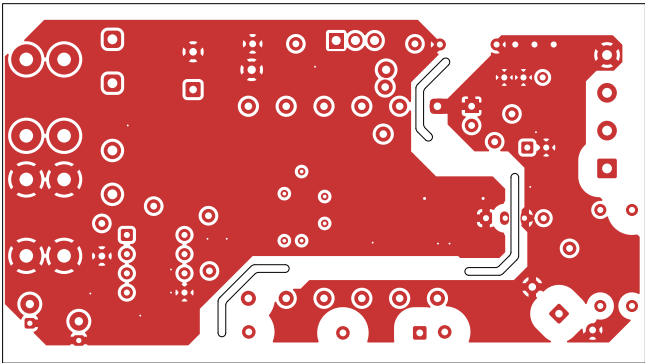


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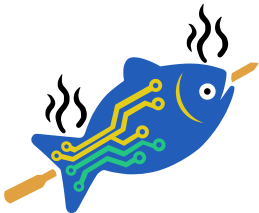
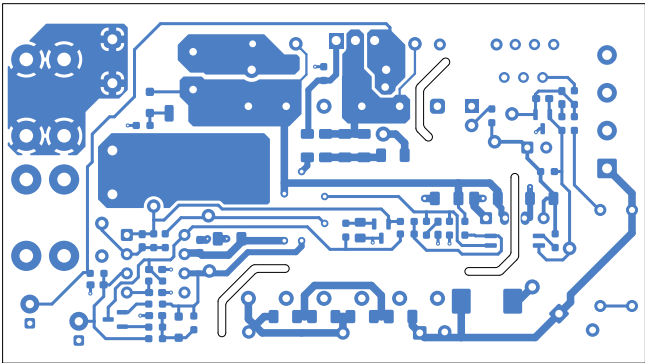
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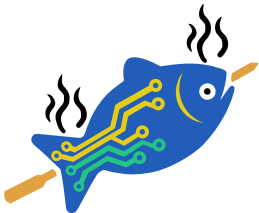
F.Cu (Scale 1:1)

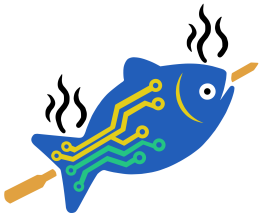
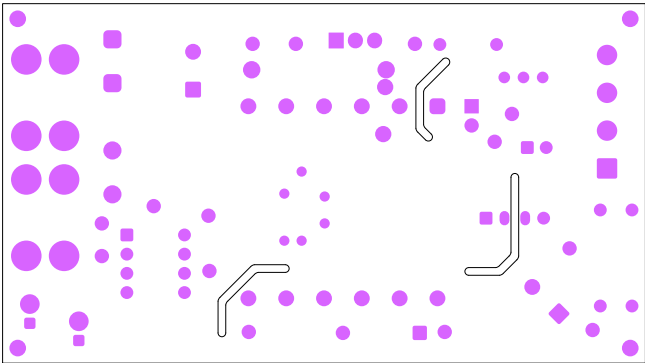


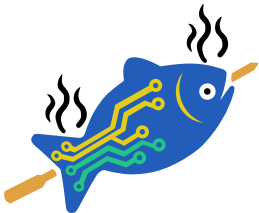
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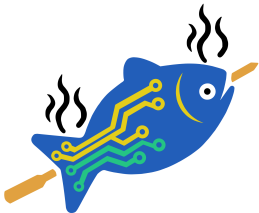
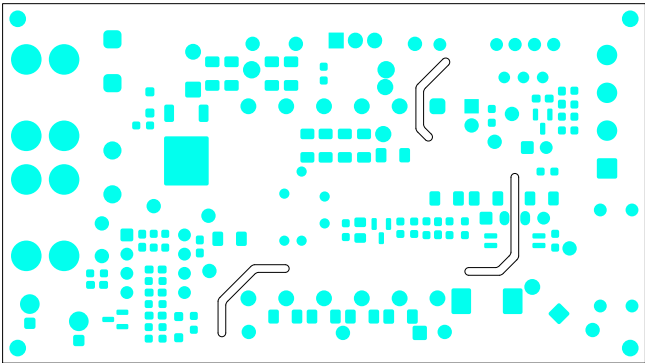
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	Step-Up		Isolated step-up 12:450 V	
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F.Cu (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	3.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size:	Sheet:
		FR	A4	5 of 10



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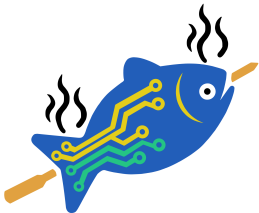
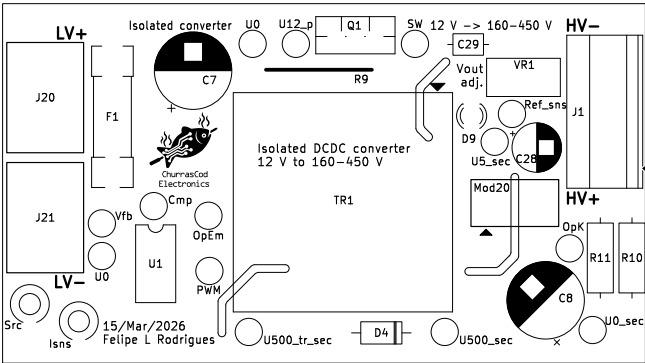


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Comments:

Company:

NA

Variant:

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Git Hash:

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Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

F.Silkscreen (Scale 1:1)

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Date:

2026-02-07

Revision:

3.0.0+ (Unreleased)

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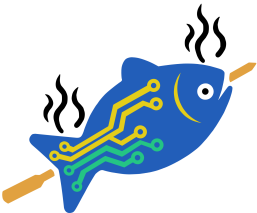
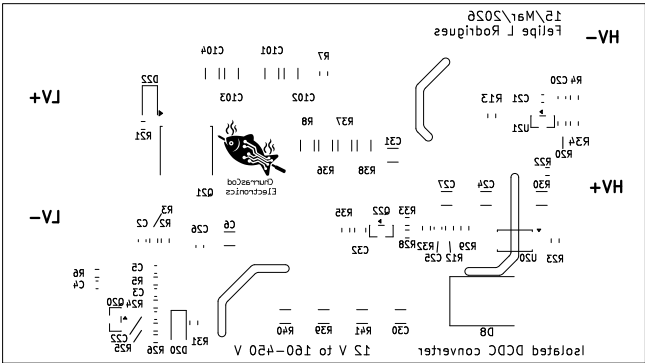
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A4

Sheet:

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Sheet Title: B.Silkscreen (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.0+ (Unreleased)
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